

Group Assignment: Time Series Analysis and Modeling using Machine Learning Models

The goal of this group project is to analyze time series dataset assigned in assessment 1, identify its underlying patterns, and apply the most appropriate machine learning model to forecast future values.

Your group must systematically analyze the assigned dataset and develop a robust predictive model. Every analytical decision made during this process must be clearly described and justified in your final report.

Please follow these specific steps:

1. Previous Descriptive Analysis and Results

- Briefly describe the dataset (from previous assessment) and previous findings (using statistical models)

2. Data Preprocessing

- Process the data as necessary to use machine learning methods. Justify why each preprocessing step was chosen.

3. Model Selection and Fitting

- Evaluate different machine learning models to determine the best fit for your specific dataset.
- Outline your selection criteria and explain the rationale behind choosing your final model over the alternatives.

4. Model Validation and Predictive Quality

- Evaluate the forecasting performance of your model using appropriate metrics and validation scheme.

Your final report can be written in a Jupyter Notebook format and must contain the Python code, details on the methodology, and the rationale behind steps 1 through 4.